Modeling Conditional Gene Dependencies in Breast Cancer through Sparse Precision Matrix Estimation

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Outline

Data and Objective

- 508 breast cancer patients
- 22,283 gene expression levels per patient
- Outcomes:
  - 105: pathological complete response (pCR)
  - 403: residual disease (RD)

Clinical Question

Can we predict **pCR vs RD** before surgery using gene expression?

- Estimate a **sparse inverse covariance** + **LDA**

Linear Discriminant Analysis (LDA)

Generative model

$$X_1, \ldots, X_{n_1} \overset{\text{iid}}{\sim} \mathcal{N}(\mu_1, \Sigma), \quad Y_1, \ldots, Y_{n_2} \overset{\text{iid}}{\sim} \mathcal{N}(\mu_2, \Sigma)$$

Common covariance Σ, class means μ_1, μ_2, priors π_1, π_2.

New point x^*:

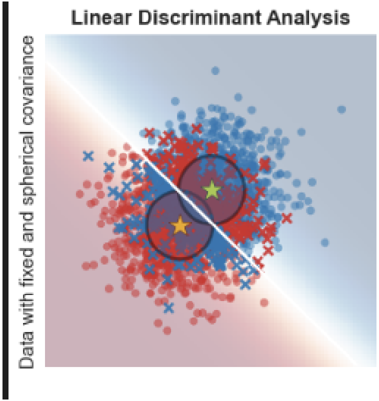
- Estimate $\hat{\mu}_1, \hat{\mu}_2, \hat{\Sigma}$, set $\hat{\Omega} = \hat{\Sigma}^{-1}$
- Compute class scores

LDA discriminant function

For class $k \in \{1, 2\}$,

$$\delta_k(x) = x^\mathsf{T} \hat{\Omega}\, \hat{\mu}_k - \frac{1}{2} \hat{\mu}_k^\mathsf{T} \hat{\Omega}\, \hat{\mu}_k + \log \hat{\pi}_k$$

Classification: assign x^* to the class with larger $\delta_k(x^*)$.



Example LDA separation in 2D feature space (illustration from online source).

LDA Contd.

Posterior Probability

of class k given $X = x$

$$\Pr(Y = k \mid X = x) = \frac{\pi_k f_k(x)}{\sum_{l=1}^{K} \pi_l f_l(x)}$$

Class prior in our setting

$$\widehat{\pi_k} = \frac{n_k}{n}$$

where n_k is the number of training samples in class k, and $n = n_1 + n_2$ is the total sample size.

Precision Matrix Estimation - Graphical Lasso (Glasso)

Penalized likelihood estimator

$$\hat{\Omega}_{\mathsf{Glasso}} = \arg\min_{\Omega \succ 0} \left\{ \langle \Omega, \Sigma_n \rangle - \log\det(\Omega) + \lambda_n \|\Omega\|_1 \right\},$$

where $\langle A, B \rangle = \mathrm{tr}(A^\mathsf{T} B)$.

KKT condition

$$\hat{\Omega}_{\mathsf{Glasso}}^{-1} - \Sigma_n = \lambda_n \hat{Z}, \quad \hat{Z} \in \partial \|\hat{\Omega}_{\mathsf{Glasso}}\|_1.$$

Equivalent constrained formulation

$$\min_{\Omega \in \mathbb{R}^{p \times p}} \|\Omega\|_1 \quad \text{s.t.} \quad \|\Omega^{-1} - \Sigma_n\|_\infty \leq \lambda_n, \ \Omega \succ 0.$$

Computationally Inefficient

- The feasible set $\left\{\Omega \succ 0 : \; \| \Omega^{-1} - S \|_\infty \leq \lambda_n\right\}$ is not separable column–wise. The $\log\det\Omega$ term couples all entries of Ω, so the optimization is globally constrained.
- Every iteration requires Ω inverting Schur complements of block matrices, can get numerically unstable or convergence can stall.
- Stringent assumptions required for exact support recovery.

CLIME Relaxation: Motivation

CLIME estimator (matrix form)

$$\hat{\Omega}_{\mathsf{CLIME}} = \arg\min_{\Omega \in \mathbb{R}^{p \times p}} \|\Omega\|_1 \quad \text{s.t.} \quad \|\Sigma_n \Omega - I\|_\infty \le \lambda_n.$$

Column-wise decomposition

For each column $i = 1, \ldots, p$,

$$\hat{\beta}_i = \arg\min_{\beta \in \mathbb{R}^p} \|\beta\|_1 \quad \text{s.t.} \quad \|\Sigma_n \beta - e_i\|_\infty \le \lambda_n,$$

where e_i is the i-th standard basis vector. Then, assemble $\hat{\Omega}$ from columns $\hat{\beta}_1, \ldots, \hat{\beta}_p$.

This is basically LP

$$\min_{\beta \in \mathbb{R}^p} \|\beta\|_1 \quad \text{s.t.} \quad \|\Sigma_n \beta - e_i\|_\infty \leq \lambda_n.$$

Relaxation via auxiliary variables: introduce $u_j \geq 0$ so that $|\beta_j| \leq u_j$. Then

$$\min_{\beta, u} \sum_{j=1}^{p} u_j$$

subject to

$$-u_j \leq \beta_j \leq u_j \quad \text{for } j = 1, \ldots, p,$$

$$\|\Sigma_n \beta - e_i\|_\infty \leq \lambda_n \quad \Longleftrightarrow \quad -\lambda_n \leq \hat{\sigma}_k^\mathsf{T} \beta - \mathbf{1}\{k = i\} \leq \lambda_n \text{ for } k = 1, \ldots, p,$$

which can be written as the pair of linear inequalities:

$$-\hat{\sigma}_k^\mathsf{T} \beta - \mathbf{1}\{k = i\} \leq \lambda_n, \quad +\hat{\sigma}_k^\mathsf{T} \beta - \mathbf{1}\{k = i\} \leq \lambda_n.$$

All constraints and the objective are linear in (β, u) $\Rightarrow$ this is a standard **LP**.

Assumptions Tradeoff?

Notation

$$\Sigma_n = (\hat{\sigma}_{ij}) = (\hat{\sigma}_1, \ldots, \hat{\sigma}_p), \quad \Sigma_0 = (\sigma^0_{ij}), \quad \mathbb{E}X = (\mu_1, \ldots, \mu_p).$$

(C1) Exponential-type tails

- There exists $0 < \eta < \frac{1}{4}$ such that

$$\frac{\log p}{n} \leq \eta.$$

- For all $i = 1, \ldots, p$ and all $|t| \leq \eta$,

$$\mathbb{E}\exp\big(t(X_i - \mu_i)^2\big) \leq K < \infty.$$

(C2) Polynomial-type tails

- For some $\gamma, c_1 > 0$,

$$p \leq c_1 n^\gamma.$$

- For some $\delta > 0$ and all $i = 1, \ldots, p$,

$$\mathbb{E}|X_i - \mu_i|^{4\gamma+4+\delta} \leq K.$$

Assumptions for ℓ_2 Error: CLIME vs Graphical Lasso

CLIME

Parameter space: approximate sparsity

$$\Omega_0 \in U(q, s_0(p)),$$

where

$$U(q, s_0(p)) := \Big\{\Omega \succ 0 : \|\Omega\|_{L_1} \le M, \quad \max_{1 \le i \le p} \sum_{j=1}^{p} |\omega_{ij}|^q \le s_0(p)\Big\},$$

for some $0 \le q < 1$, $M > 0$.

Additional assumptions:

- Eigenvalues of Ω_0 bounded (well-conditioned).

Graphical Lasso (Glasso)

Structural assumptions:

- Exact sparsity of $\Omega^\star$ on support S.
- Eigenvalues of $\Sigma^\star$ bounded away from 0 and ∞.

Incoherence / irrepresentability: Let $\Sigma^\star = (\Omega^\star)^{-1}$ and $\Gamma^\star = \Sigma^\star \otimes \Sigma^\star$. Partition by support S vs S^c:

$$\left\|\Gamma^\star_{S^c S}(\Gamma^\star_{SS})^{-1}\right\|_\infty \le 1 - \alpha, \qquad \alpha \in (0, 1).$$

[a]

[a] Intuitively, zero entry of $\Omega^\star$ cannot be too strongly correlated (in the Fisher-information sense) with the collection of nonzero entries. Formally, every row of $\Gamma^\star_{S^c S}(\Gamma^\star_{SS})^{-1}$ must have ℓ_1-norm at most $1 - \alpha$, i.e. strictly bounded away from 1.

Performance Guarantees

Assume $\Omega_0 \in U(q, s_0(p))$ and choose $\lambda_n \asymp M\sqrt{\frac{\log p}{n}}$.

(C1) Exponential-type tails

$\lambda_n = C_0 M\sqrt{\frac{\log p}{n}}$ with C_0 depending on (η, τ, K), then with probability at least $1 - 4p^{-\tau}$,

$$\|\hat{\Omega} - \Omega_0\|_2 \;\lesssim\; M^{2-2q}\, s_0(p) \left(\frac{\log p}{n}\right)^{(1-q)/2}.$$

(C2) Polynomial-type tails

$\lambda_n = C_2 M\sqrt{\frac{\log p}{n}}$ with C_2 depending on (τ, θ), then with probability at least $1 - O\big(n^{-\delta/8} + p^{-\tau/2}\big)$,

$$\|\hat{\Omega} - \Omega_0\|_2 \;\lesssim\; M^{2-2q}\, s_0(p) \left(\frac{\log p}{n}\right)^{(1-q)/2}.$$

Convergence rate better in some regime

- For ℓ_1–MLE type estimators, the convergence rates in the case of polynomial–type tails are typically much slower than those in the case of exponential–type tails (see, e.g., Ravikumar et al. 2008).
- CLIME attains the same rates of convergence under either of the two moment conditions and significantly outperforms ℓ_1–MLE type estimators in the case of polynomial–type tails.

Competing Benchmark: Nodewise Regression

Gaussian conditional representation. If $X \sim \mathcal{N}(0, \Sigma^\star)$, then for each node $j \in \{1, \dots, p\}$,

$$\mathbb{E}(X_j \mid X_{-j}) = \sum_{i \neq j} \beta^\star_{ij} X_i, \qquad \beta^\star_{ij} = -\frac{\Omega^\star_{ij}}{\Omega^\star_{jj}}, \qquad \mathrm{Var}(X_j \mid X_{-j}) = \frac{1}{\Omega^\star_{jj}}.$$

Nodewise Lasso regressions. For each $j \in \{1, \dots, p\}$, solve

$$\widehat{\beta}^{(j)} \in \arg\min_{\beta \in \mathbb{R}^{p-1}} \left\{ \frac{1}{2n} \|X_j - X_{-j}\beta\|_2^2 + \lambda_j \|\beta\|_1 \right\},$$

where $\lambda_j \asymp \sqrt{(\log p)/n}$.

Residual variance and assembly. Let $\widehat{\sigma}_j^2 = (1/n)\, \|X_j - X_{-j}\widehat{\beta}^{(j)}\|_2^2$. Define

$$\widehat{\Omega}_{jj} = \frac{1}{\widehat{\sigma}_j^2}, \qquad \widehat{\Omega}_{ij} = -\,\widehat{\Omega}_{jj}\, \widehat{\beta}_i^{(j)}, \quad i \neq j,$$

which plugs in $\Omega^\star_{ij} = -\Omega^\star_{jj}\beta^\star_{ij}$ and $\Omega^\star_{jj} = 1/\,\mathrm{Var}(X_j \mid X_{-j})$.

Competing Benchmarks contd.

SCAD-penalized estimator

$$\hat{\Omega}_{\mathsf{SCAD}} := \arg\min_{\Omega \succ 0} \left\{ \langle \Omega, \Sigma_n \rangle - \log\det(\Omega) + \sum_{i=1}^{p} \sum_{j=1}^{p} \mathsf{SCAD}_{\lambda,a}(|\omega_{ij}|) \right\}.$$

SCAD penalty (Fan & Li)

Let $t \geq 0$, $\lambda > 0$, $a > 2$. The derivative of the SCAD penalty is

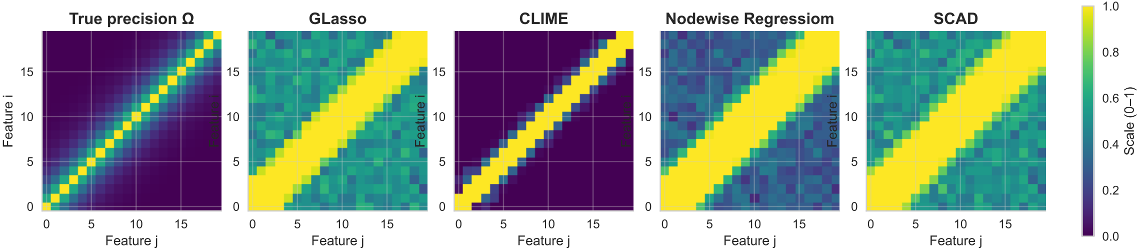
$$\mathsf{SCAD}_{\lambda,a}(t) = \begin{cases} \lambda, & 0 \leq t \leq \lambda, \\ \dfrac{a\lambda - t}{a-1}, & \lambda < t \leq a\lambda, \\ 0, & t > a\lambda. \end{cases}$$

Simulations

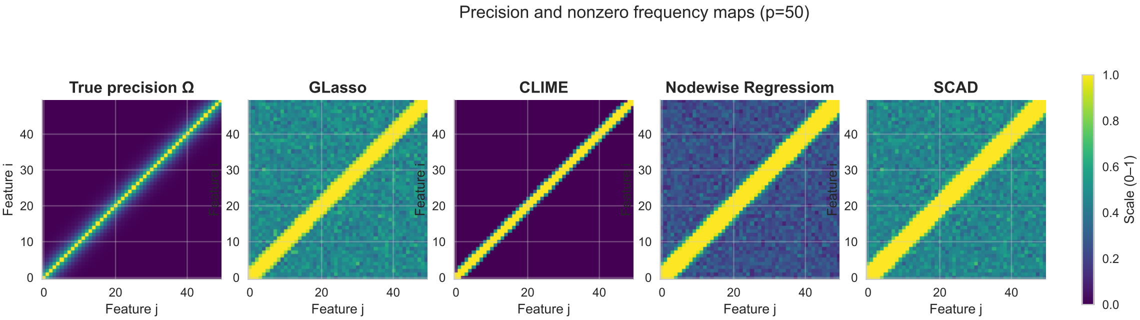
Model 1 (Toeplitz / AR(1)-type):

$$\Omega^0_{ij} = 0.6^{\,|i-j|}, \qquad 1 \le i, j \le p.$$

Precision and nonzero frequency maps (p=20)



Simulations: Toeplitz Contd.



Explanation on simulations

- Every off-diagonal is nonzero, so exact sparsity is violated ($s_0(p) = p$). Glasso work poor.
- Nodewise regression: each node is conditionally dependent on all others, with coefficients decaying like $\rho^{|i-j|}$, works slightly better than Glasso.
- CLIME assumes approximate sparsity via

$$\Omega_0 \in U(q, s_0(p)), \qquad \max_{1\le i\le p} \sum_{j=1}^{p} |\omega_{ij}|^{\,q} \le s_0(p), \qquad 0 \le q < 1.$$

In this model,

$$\sum_{j=1}^{p} \left|\Omega^0_{ij}\right|^{\,q} \;=\; \sum_{k=-\infty}^{\infty} 0.6^{\,q|k|}$$

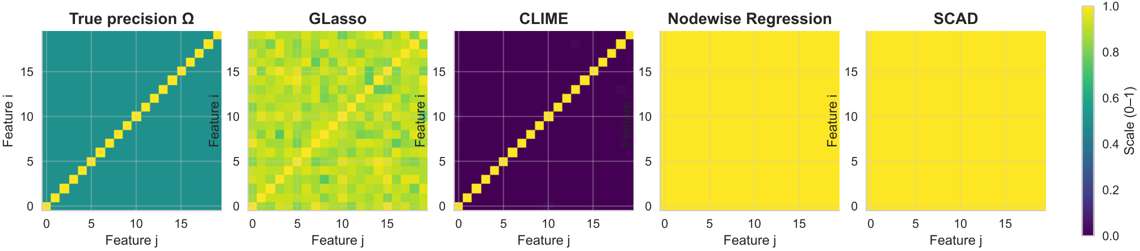
is a geometric series, so it is bounded by a constant independent of p for any fixed $q > 0$.

Simulation Model 2: Dense Precision Matrix

Model 2 (non-sparse / dense):

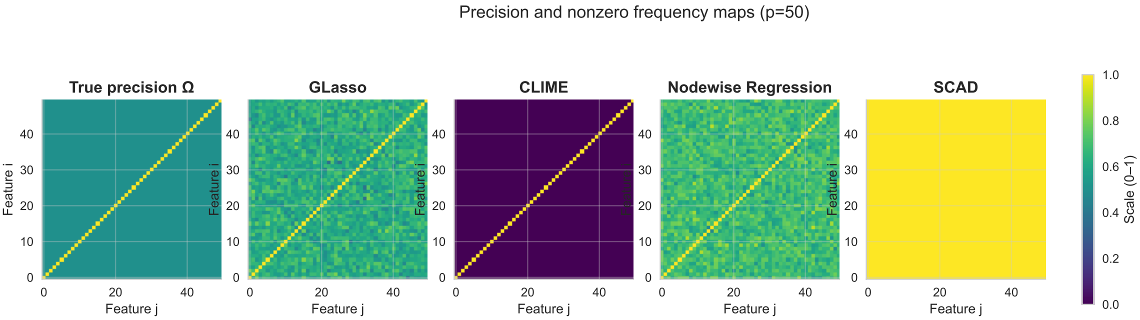
$$\Omega^0_{ij} = \begin{cases} 1, & i = j, \\ 0.5, & i \neq j, \end{cases} \qquad 1 \leq i, j \leq p.$$

Precision and nonzero frequency maps (p=20)



Dense Precision Matrix for high dimension

No sparsity at all - ideally nothing should work, Nodewise Regression outperforms.



Comparison of average (SE) l_2 loss for model 1

Table: Results for Model 1

	Model 1			
p	$\hat{\Omega}_{\mathrm{CLIME}}$	$\hat{\Omega}_{\mathrm{Glasso}}$	$\hat{\Omega}_{\mathrm{Nodewise}}$	$\hat{\Omega}_{\mathrm{SCAD}}$
20	1.71 (0.07)	1.82 (0.04)	1.83 (0.06)	3.01 (0.009)
30	2.21 (0.06)	2.37 (0.06)	2.34 (0.06)	2.97 (0.01)
40	2.32 (0.08)	2.31 (0.06)	2.30 (0.07)	3.08 (0.01)
50	3.14 (0.004)	3.23 (0.003)	2.89 (0.004)	3.18 (0.008)

Comparison of average (SE) l_2 loss for model 2

Table: Results for Model 2

		Model 2		
p	$\hat{\Omega}_{\text{CLIME}}$	$\hat{\Omega}_{\text{Glasso}}$	$\hat{\Omega}_{\text{Nodewise}}$	$\hat{\Omega}_{\text{SCAD}}$
20	21.89 (0.01)	24.96 (0.01)	18.89 (0.01)	17.99 (0.01)
30	32.01 (0.02)	28.90 (0.01)	27.73 (0.02)	39.98 (0.01)
40	47.82 (0.07)	35.03 (0.05)	34.53 (0.05)	59.82 (0.01)
50	70.33 (0.04)	60.04 (0.05)	61.22 (0.04)	84.10 (0.02)

Revisit the breast cancer problem

- **Training data:**
    - pCR: 80 subjects
    - RD: 323 subjects
    - Total $n_{\text{train}} = 403$
- **Feature selection on training set only:**
    - Start with all gene expressions.
    - For each gene, perform a two-sample t-test (pCR vs RD).
    - Select the top 500 genes (ranked by smallest p-values).

Deliberately choose a **high-dimensional** setting:

$$p = 500 \ > \ n_{\text{train}} = 403.$$

- **Test data (held out):**
    - pCR: 25 subjects
    - RD: 80 subjects

Classification Performance (Test Set)

Definitions

$$\text{Sensitivity} = \frac{\#\{\text{pCR correctly predicted}\}}{\#\{\text{all pCR}\}} = \frac{TP}{TP + FN},$$

$$\text{Specificity} = \frac{\#\{\text{RD correctly predicted}\}}{\#\{\text{all RD}\}} = \frac{TN}{TN + FP}.$$

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}.$$

Method	Sensitivity	Specificity	MCC
Glasso	0.674	0.767	0.403
Nodewise	0.721	0.792	0.477
SCAD	0.612	0.820	0.492
CLIME	0.829	0.783	0.505

Conclusion

- CLIME is computationally less heavy than the existing methods in precision matrix estimation.
- There is no estimation method uniformly better than others in this context.
- CLIME outperforms in a near / limiting sparsity a compared to Glasso.

References

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Thank You!

Questions or comments?